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RRG5004

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Gen 4 DDR5 Registering Clock Driver

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Overview

Product Options

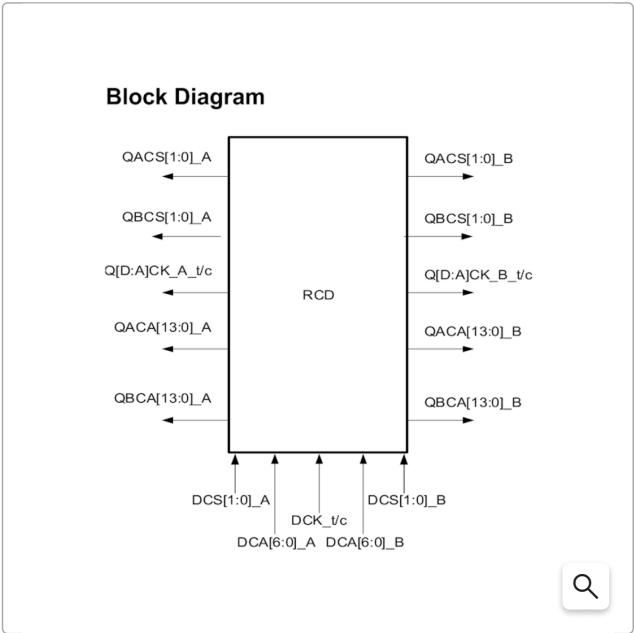
Documentation

Boards & Kits

Support

Features

- This RCD enhances the performance and reliability of memory modules by ensuring precise timing and signal integrity. Renesas RCD RRG5004 is validated by all major memory vendors.
- Compliant with JEDEC DDR5RCD04 specification
- Pinout optimized for DDR5 RDIMM PCB layout
- DDR5 server speeds up to 7200MT/s
- Supports power-down modes to conserve server power
- Supports 1-rank/2-rank DIMM configurations
- Supports single die package (SDP), dual die package (DDP), and 3D stacking DRAM types
- Provides access to internal control words for configuring device features and adapting to different RDIMM system applications
- I²C and I3C sideband access for asynchronous register access control
- Loopback and pass-through modes
- Package: 8.7 × 13.5 mm, 240-FCBGA



RRG5004x-A00 Block Diagram

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Description

The RRG5004 (Gen 4 RCD) is a registering clock driver used on DDR5 RDIMMs. It supports DDR5 server speeds up to 7200 MT/s. Its primary function is to buffer the Command Address (CA) bus, chip selects, and clock between the host controller and the DRAMs.

- **Dual-Channel Design:** Operates independently with shared common logic such as clocking, enhancing performance and reliability.
- **High-Speed Support:** Achieves speeds up to 7200 MT/s, catering to the demands of modern server environments.
- **Versatile Compatibility:** Supports both x4 and x8 DRAM configurations, providing flexibility for various applications.
- **I3C Bus Interface:** Facilitates sideband communication with the CPU or motherboard Baseboard Management Controller (BMC), ensuring smooth operations and management.
- **Efficient Data Handling:** Each channel features a 7-bit double data rate CA bus input, single parity input, two chip-select inputs, and produces two copies of 14-bit single data rate CA bus outputs and chip-select outputs.
- **Integrated Clock Management:** Common clock input and PLL with separate clock outputs for each DRAM channel, optimizing synchronization and performance.

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Parameters

Attributes	Value
Product Type	Registering Clock Driver
Function	DDR5 Gen 4.0 Server RCD
Application System	Server
Input Voltage Range (V)	1.06 – 1.16

Package Options

Pkg. Type	Pkg. Dimensions (mm)	Lead Count (#)	Pitch (mm)
FCCSP	13.5 x 8.7 x 0.9	240	0.65

Applications

Server/Data Center



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